Lecture 13 — Span, independence, and subspaces

Wed Sep 23

Last updated: September 22, 2026.

Boyd video (optional): Lecture 15 — linear independence · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Traditional vocabulary supplement to Boyd (span, subspace), building on Boyd 5.1. Related objectives: Lectures. Schedule. Midterm 1 is Friday, October 2.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Check yourself**.

Learning objectives

By the end of class, students should be able to:

1. Define span using linear combinations.
2. Determine whether a vector lies in a simple span.
3. Interpret spans geometrically as lines, planes, or higher-dimensional collections.
4. Recognize basic subspaces and distinguish them from shifted lines or planes.

Warm-up

Let $a = (1, 0)$ and $b = (0, 1)$. Can you write $v = (3, -2)$ as $\alpha a + \beta b$? If yes, give α and β.

Fill in

The **span** of vectors $a_1, \dots, a_k$ is the set of all **linear combinations**

$$\text{span}\{a_1, \dots, a_k\} = \{\, \beta_1 a_1 + \cdots + \beta_k a_k \mid \beta_i \in \mathbb{R} \,\}.$$

So v lies in the span iff the equation $\beta_1 a_1 + \cdots + \beta_k a_k = v$ has _____.

Geometry (in $\mathbb{R}^2$ or $\mathbb{R}^3$):

- Span of one nonzero vector: a _____ through the origin.
- Span of two non-parallel vectors in $\mathbb{R}^3$: a _____ through the origin.
- Span of the standard basis of $\mathbb{R}^n$: all of _____.

A **subspace** of $\mathbb{R}^n$ is a nonempty set V closed under addition and scalar multiplication (equivalently: closed under linear combinations). Every subspace contains _____.

Spans are always subspaces. A line or plane that **does not** pass through the origin is _____ a subspace (it fails to contain 0, or fails closure under scaling).

Independence vs span: an independent list that spans V will be called a **basis** of V next class. Today, focus on "can I build v from these vectors?"

Worked in class

W1. Membership. Let $a = (1, 1)$ and $b = (1, -1)$.

(a) Is $v = (2, 0)$ in $\text{span}\{a, b\}$? Find coefficients or explain why not.

(b) Is $w = (1, 0, 0)$ in $\text{span}\{(1, 1, 0), (0, 1, 1)\}$ inside $\mathbb{R}^3$?

W2. Geometry. Sketch in $\mathbb{R}^2$: $\text{span}\{(2, 1)\}$ and $\text{span}\{(1, 0), (0, 1)\}$.

W3. Subspace check. Which of the following are subspaces of $\mathbb{R}^2$? Brief reason.

(a) All vectors (x, y) with $y = 3x$.

(b) All vectors (x, y) with $y = 3x + 1$.

(c) The single point $\{(0, 0)\}$.

(d) All vectors with $\|(x, y)\| = 1$ (the unit circle).

W4. Dependent generators. Let $a = (1, 0)$, $b = (2, 0)$, $c = (0, 1)$. Explain why $\text{span}\{a, b, c\} = \text{span}\{a, c\}$. What does that say about redundant generators?

W5. Application. Two feature directions f_1 ="morning temperature pattern" and f_2 ="afternoon pattern" as 24-vectors. What does it mean if a third pattern g lies in $\text{span}\{f_1, f_2\}$?

Check yourself

C1. Complete: $v \in \text{span}\{a, b\}$ means $v =$ _____ for some scalars.

C2. Is $(3, 3)$ in $\text{span}\{(1, 1)\}$? Coefficients if yes.

C3. True or false: every subspace of $\mathbb{R}^n$ is a span of some list of vectors.

C4. Why is the set of vectors with first entry equal to 1 not a subspace of $\mathbb{R}^2$?

C5. Give a vector in $\mathbb{R}^2$ that is **not** in $\text{span}\{(1, 2)\}$.

Answers (Check yourself)

C1. $v = \alpha a + \beta b$ (any equivalent wording).

C2. **Yes.** $(3, 3) = 3 \cdot (1, 1)$.

C3. **True** (in $\mathbb{R}^n$ every subspace has a basis, hence is a span). For this course it is enough to know that the subspaces we meet are spans.

C4. It does not contain 0 (first entry would be 0), or equivalently it is not closed under scaling.

C5. Example: $(1, 0)$, or $(2, 1)$, or any vector not a multiple of $(1, 2)$.